(a) (10 points) Complete the function **policyIteration**. The iterations should use synchronous sweeps, and a sufficiently large number of iterations in the policy evaluation part to guarantee that v_π is practically achieved. Then use **policyIteration** to find an optimal policy and value function for **SmallGrid**. For how long you need to apply the policy evaluation and policy improvement steps until convergence with $\gamma = 0.9$ and $\eta = 0.2$? *Note:* Count the number of times you cycle over policy evaluation and policy improvement.

For part a, it took 106 policy evaluations and 4 policy improvements.

(b) (10 points) Complete the function **valueIteration**. The iterations should use synchronous sweeps. Then use **valueIteration** to find an optimal policy and value function for **SmallGrid**. For how long you need to iterate achieve convergence error $\epsilon = 1\text{e}{-3}$ with $\gamma = 0.9$ and $\eta = 0.2$? Does the number of iterations change for different value function initializations? Is the policy the same as in part (a)?

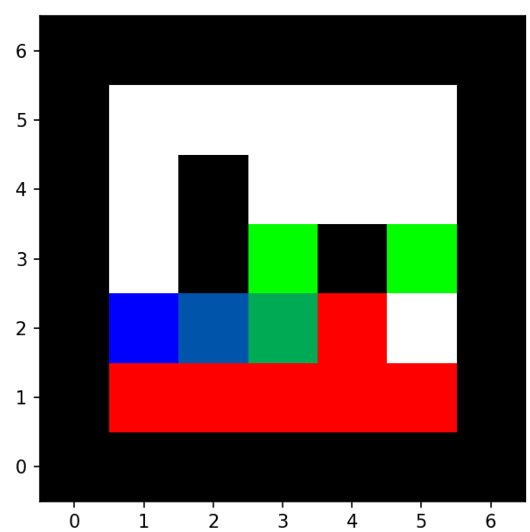
Table 1: Value Iteration Convergence for Different Initializations

Initialization $V_0(s)$	Iterations to converge
0 for all states	66 iterations
1 for all states	65 iterations
-5 for all states	70 iterations
10 for all states	47 iterations

Different initializations change the number of iterations because the initial value estimate may be closer or farther from the optimal value function. However, value iteration still converges to the same optimal policy and value function due to the contraction property of the Bellman optimality operator.

(c) (5 points) Using **MediumGrid** with the default cost function, play with the discount factor γ and the noise parameter η to obtain different results in this environment. Can you find parameters such that your optimal policy from value iteration goes through the cell next to the start cell, given by (2,2)?

Using $\gamma = 0.5$ and $\eta = 0.1$. We get the following optimal path [(2, 2), (3, 2), (3, 3)] starting from (1,2) as seen below

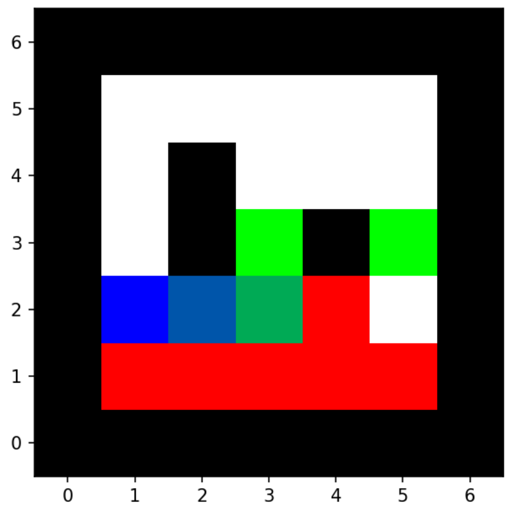


(d) (15 points) Consider the `MediumGridBridge` layout. This grid has the same layout as `MediumGrid`, but has two special (goal) states with positive payoff at $(5, 3)$ and at $(3, 3)$. The bottom row of the environment has states with negative payoff of -10 (cliff states). The starting state is at $(1, 2)$. We call the state $(2, 2)$ the "bridge" between the start and the cliff states. We distinguish between two types of paths: (1) paths that "risk the cliff" and travel near the bottom row of the grid; these paths are shorter but risk earning a large negative payoff. (2) Paths that "avoid the cliff" and travel along the top edge of the grid. These paths are longer but are less likely to incur huge negative payoffs (or large costs.)

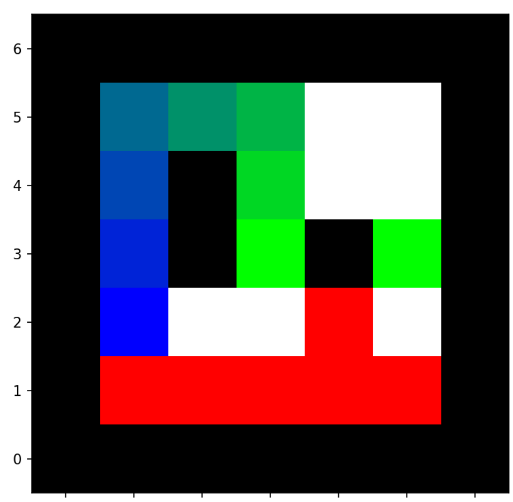
Give an assignment of parameter values for the discount γ, and noise value η in the actions which produce an optimal policy that results in the following paths:

1) Prefer the close exit (with cost -1), risking the cliff (with cost +10)
2) Prefer the close exit (with cost -1), but avoiding the cliff (with cost +10)
3) Prefer the distant exit (with cost -10), risking the cliff (with cost +10)
4) Prefer the distant exit (with cost -10), but avoiding the cliff (with cost +10)

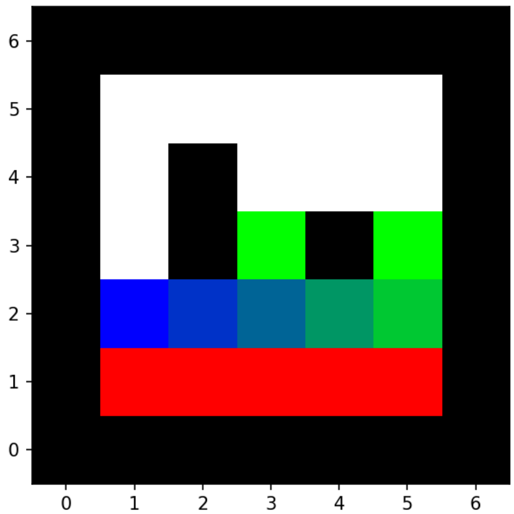
Part (1): $\gamma = 0.4$, and $\eta = 0.02$
[(2, 2), (3, 2), (3, 3)]



Part (2): $\gamma = 0.3$, and $\eta = 0.2$
[(1, 3), (1, 4), (1, 5), (2, 5), (3, 5), (3, 4), (3, 3)]



Part (3): $\gamma = 0.9$, and $\eta = 0.2$
[(2, 2), (3, 2), (4, 2), (5, 2), (5, 3)]



Part (4): $\gamma = 0.9$, and $\eta = 0.5$
[(1, 3), (1, 4), (1, 5), (2, 5), (3, 5), (4, 5), (5, 5), (5, 4), (5, 3)]